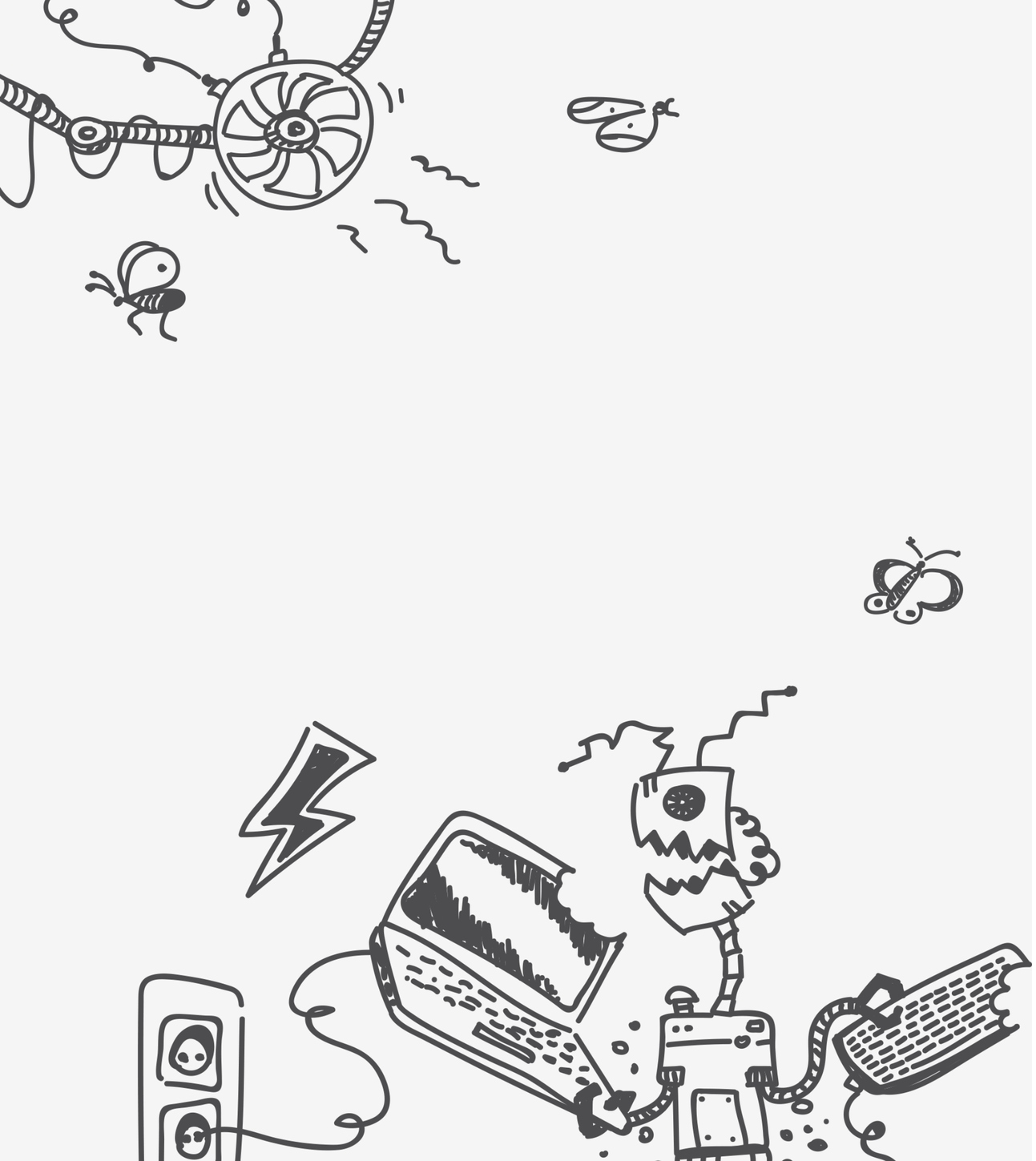

Framing the problem

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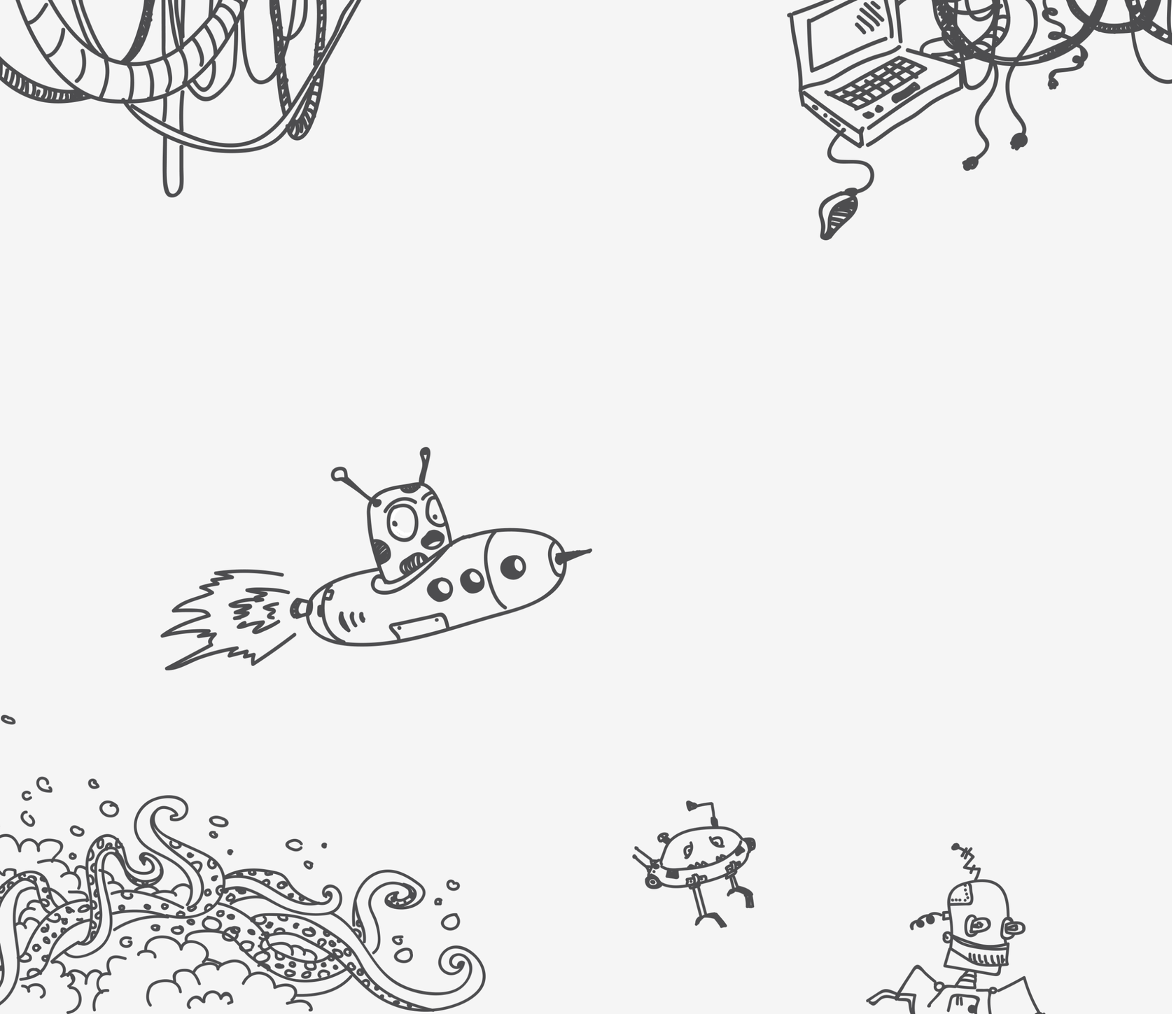
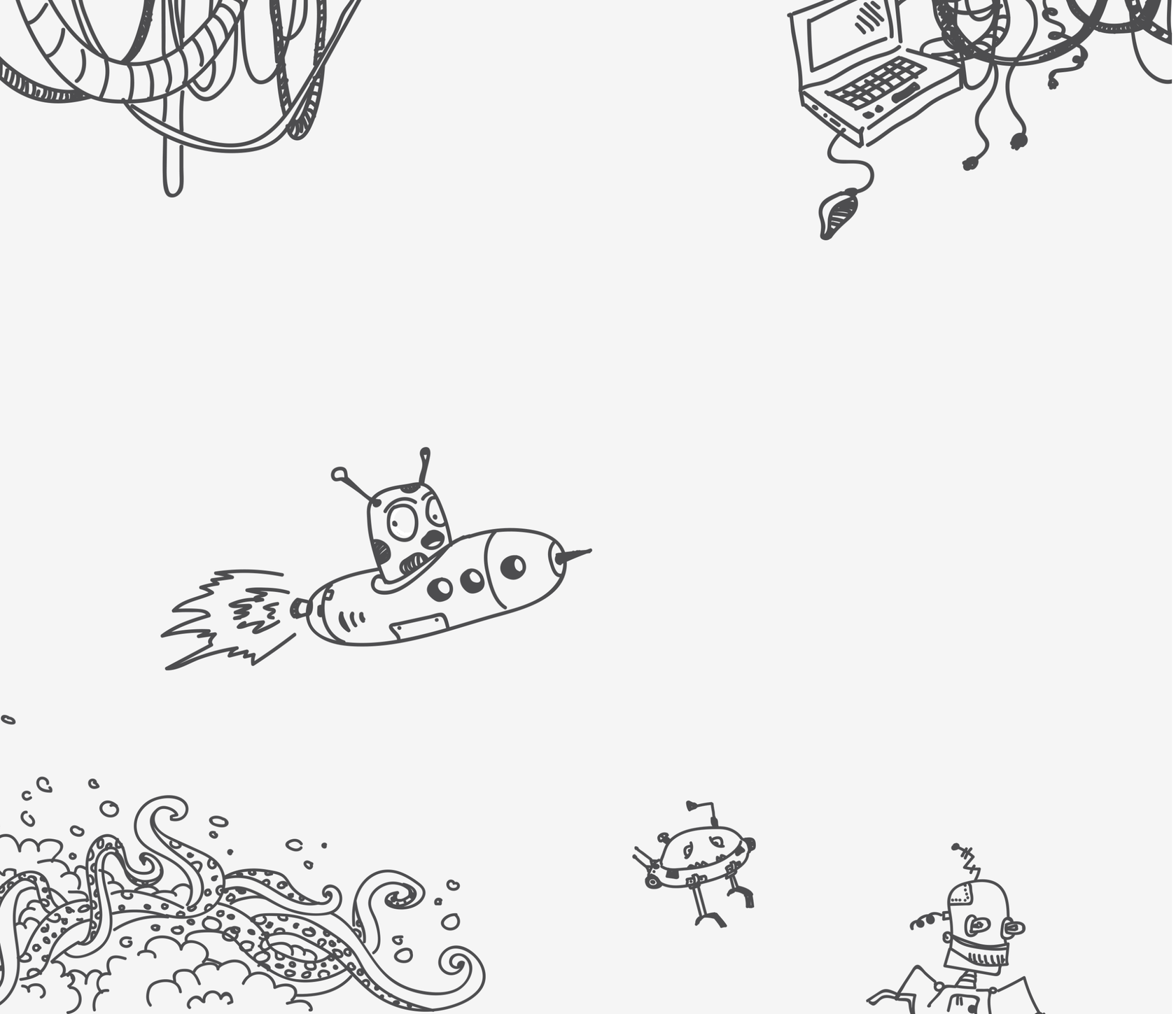


Introduction

Introduction

During the prework, we conducted our first data analysis together.

Data analysis is a creative process when data becomes useful knowledge that can be used to make better decisions. The first step of the Data Science process (that we chose to work with) is to define the problem.



Formulating questions as the best tool used to understand the problem

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Formulating questions

Let's consider a question for a moment. What is the best tool available to any of us who needs to understand any issue?

QUESTIONS.

It is said that "one who asks questions doesn't lose their way," and this also proves true during data analysis. Asking a large number of questions will certainly dramatically increase the knowledge of the problem you're analyzing. The exact way you do this will depend on the context in which you're working.

One thing remains constant - whether you work in the public or private sector, or perhaps you're an employee of a university - before you can begin to solve a problem, you must understand it in depth.

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Understanding the problem

In order to understand your customers' problem (most often these will be people directly related to the business part of your organization), it's a good idea to put yourself in their shoes. This makes it easier to understand their view of the problem and develop the desired solution. E.g., in a large company, the process will begin by talking to product owners, managers and supervisors to determine business priorities and strategic decisions.

Often, the originally formulated queries you receive as a data analyst don't tell you much and are ambiguous.

Understanding the problem

Suppose you work for a video production company that creates for an online video on demand (VOD) platform. The sales manager comes to you, announcing that the company wants to increase subscription sales. For you as a data analyst, this isn't entirely clear, and your job will be to translate this task into a specifically defined problem that can be solved using data. E.g., predicting the conversion rate and ROI of different customer groups.

Domain knowledge and product intuition are most important in this step. You will gain both of these skills over time by talking to subject matter experts. Domain knowledge will allow you to understand the terminology associated with the issue. On the other hand, learning about the product or service will be very helpful in terms of the intuition behind the metrics.

Scale of the problem - does it affect the data analysis process?

Typically, as humans, we often value what is around us. E.g., a computer can be cheap or expensive. Likewise, the problem we have to solve as analysts can be easy, typical or difficult.

The scale of the problem often depends on:

- The size and variety of the data being analyzed,
- the size of our organization,
- the number of people involved in the product/process being analyzed,
- on the value of the problem itself.

In the small scale, the problem is often easy to define. In the large scale, e.g., in a big organization, the problem may be completely unrelated to our typical work. It may concern people, products or processes previously unknown to us. It doesn't always have to directly translate into revenue, costs, profit or other core company goals. It is very important to understand all the relationships and formulate the problem in the context of the data.

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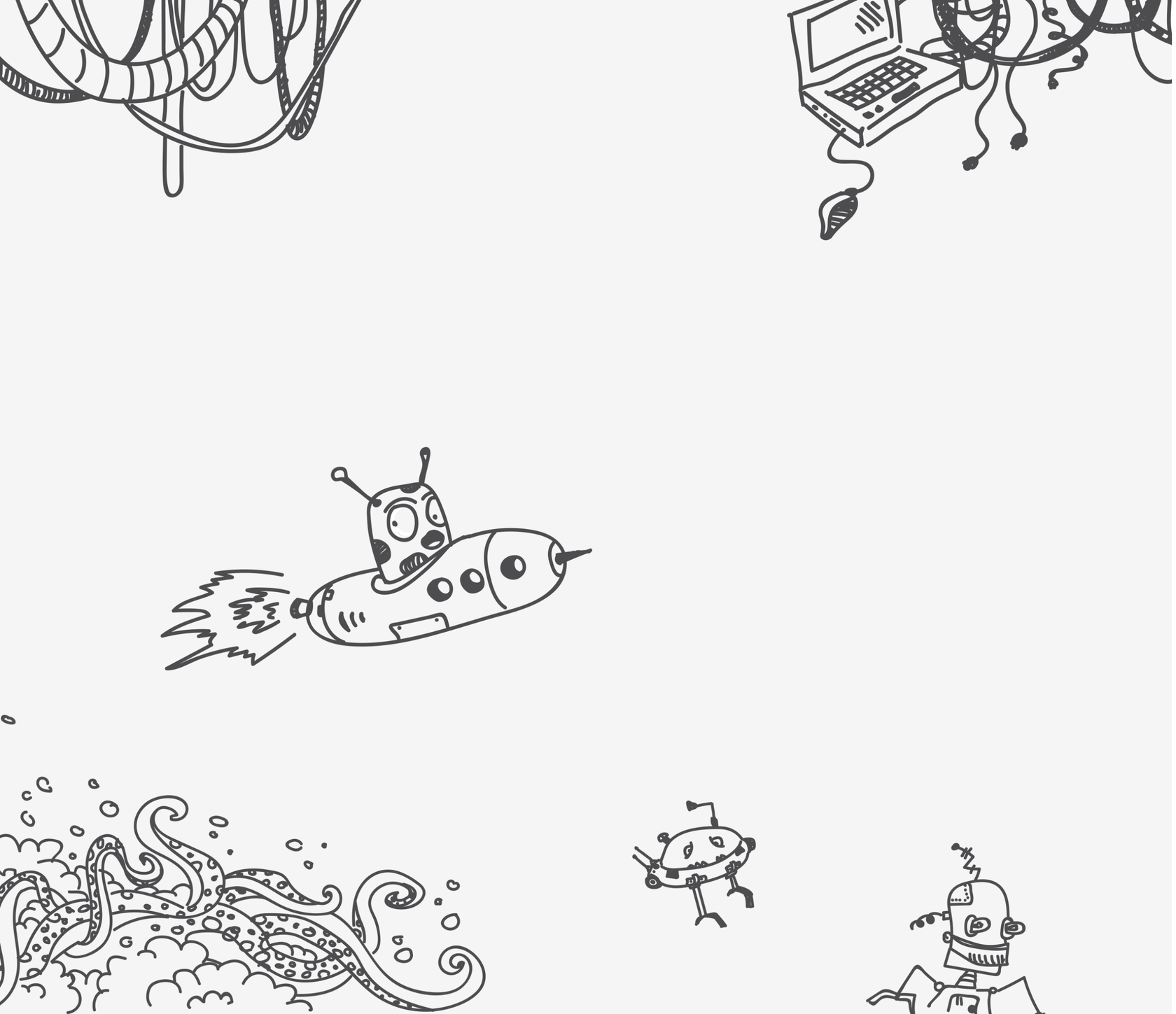
Tools to help define and understand the problem

To summarize the theory - the tools that will help you define the problem are:

- the ability to systematically acquire knowledge of the problem to be solved,
- methods for analyzing the performance of a product (both existing and yet to be planned),
- the ability to understand the company's business strategy,
- The ability to gather the team and resources (stakeholders' time), data and access to the relevant systems (e.g. the application whose data you're going to analyze).



Example of understanding and framing the problem

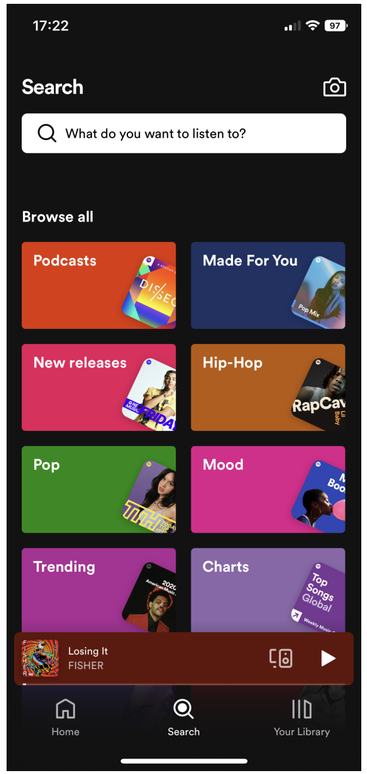
Example of understanding and framing the problem

Without a theoretical basis, it would be practically impossible to proceed to the solution of any task. Nevertheless, it is said that "practice makes perfect", so let's proceed to apply our theory in practice. During the course of the classes we'll deal with data analysis in a field that is certainly close to everyone's heart.

We are talking about **music** - that is, a fine art that has an extraordinary impact on human beings. Music allows you to motivate yourself (e.g. during a hard workout), relax (e.g. after work), and improve your mood. It also turns out that selected genres of music (such as relaxation music) have an impact on your health, as they allow us to increase relaxation, so that our heart rate lowers and the level of the stress hormone (cortisol) decreases.

Ways of listening to music

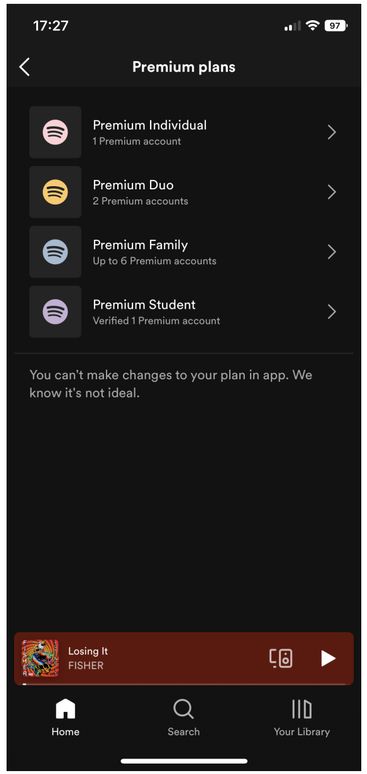
The way people listen to music has changed dramatically over the years - from turntables to cassette players, radio or CDs to streaming services. Of course, we can also listen to music live, at concerts and festivals, which has its own unique atmosphere. But let's go back to the media, which allow us to listen to music literally anywhere. In recent years, streaming services have captured a huge share of the music market. One of the most popular streaming services is Spotify.



Spotify - short description

Spotify is a streaming service founded in Sweden that offers access to music and podcasts. It offers more than 82 million songs created by artists from around the world. The service operates under a "freemium" license, meaning that basic functionality is available free of charge, but it has its limitations in the form of ads and less app personalization options.

Additional features, such as the ability to download music directly to a device (and offline listening) or ad-free listening, are available with paid subscriptions.



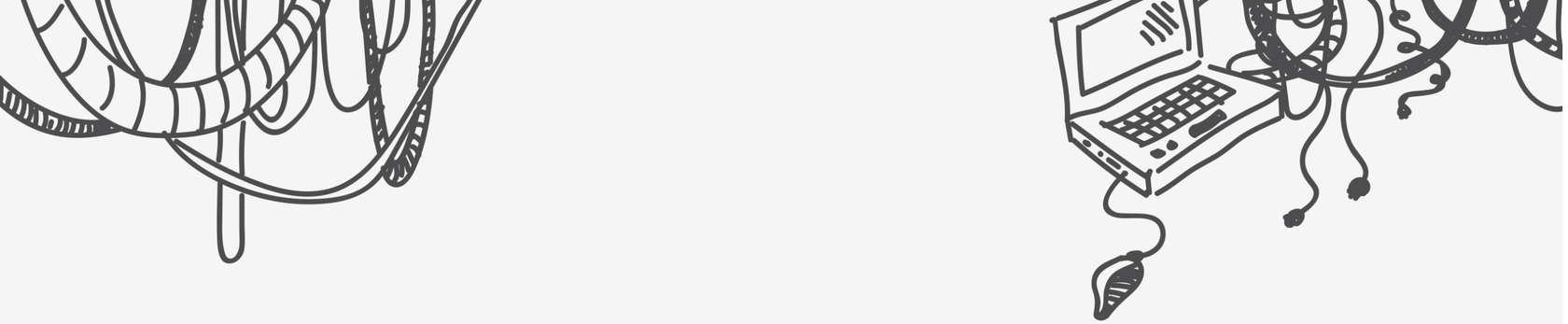
Spotify - short description

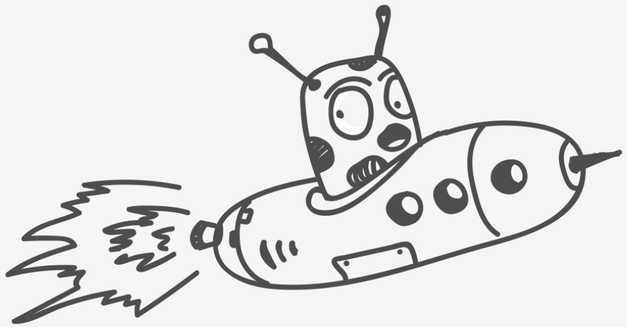
Users can search for music based on a number of parameters such as:

- song title,
- artist,
- album,
- genre.

The application also allows us to create our own lists of favorite songs, which we can share with friends.

According to a report that Spotify as a publicly traded company released after the second quarter of 2022, the number of users with an account on the service is as high as 433 million! Of the total users, 188 million use a paid subscription.





Aim of the analysis

Aim of the analysis

Requirements

We have been asked to prepare an analysis that answers the following questions:

1. Which artists have the most songs listed as the most popular?
2. Is there a correlation between the duration of a song and the year the song was released?
3. Do features such as danceability, energy or loudness influence the popularity of the song?
4. Among the performers listed as having the most popular songs, can we find those whose works include mostly negative songs?
5. Which artists have the most songs listed among the most popular songs?
6. Does average song tempo change based on the release year?

Aim of the analysis

Understanding the data at our disposal

The person responsible at Spotify for the product of playlists with the most popular songs asked us to perform a data analysis based on nearly 2000 of the most popular songs on Spotify between 2000 and 2019. So let's take a look at the definition of the data we were given to better understand it and, consequently, better understand the task at hand, which we'll discuss in a moment.

- **artist_id** - id of the artist,
- **artist** - name of the artist (or surname, pseudonym),
- **song** - title of the song,
- **duration_ms** - song duration in milliseconds,
- **explicit** - indicates whether the lyrics are offensive or inappropriate for kids,
- **year_id** - identifier of release year,
- **year** - release year
- **popularity** - higher values indicate greater song popularity

Aim of the analysis

Understanding the data at our disposal

- **danceability** - describes how suitable a song is for dancing based on elements such as tempo, rhythm stability, beat strength and overall regularity. A value of 0 means that the song isn't suitable for dancing, a value of 1 indicates that the song is very danceable
- **energy** - represents a perceptual measure of intensity and activity. Higher values indicate more energetic songs.
- **key** - The key in which the track is located. Integers refer to pitch using standard pitch class notation. E.g., 0 = C, 1 = C♯/D♭, 2 = D and so on. If no key is detected, the value is -1
- **loudness** - the overall volume of the track in decibels. Loudness values are averaged across the entire song and are useful for comparing the relative loudness of songs. Loudness is the quality of sound, being the primary psychological correlate of physical strength (amplitude)
- **mode** - indicates the modality (major or minor) of the track, specifying the type of scale from which the melodic content is derived
- **speechiness** - states whether the lyrics contain spoken words. A value closer to 1 indicates predominantly spoken content (e.g., audiobooks or podcasts). Values between 0.33 and 0.66 can indicate a mix of speech and music (e.g., rap tracks with certain characteristics). Below 0.33, music tracks are primarily found.

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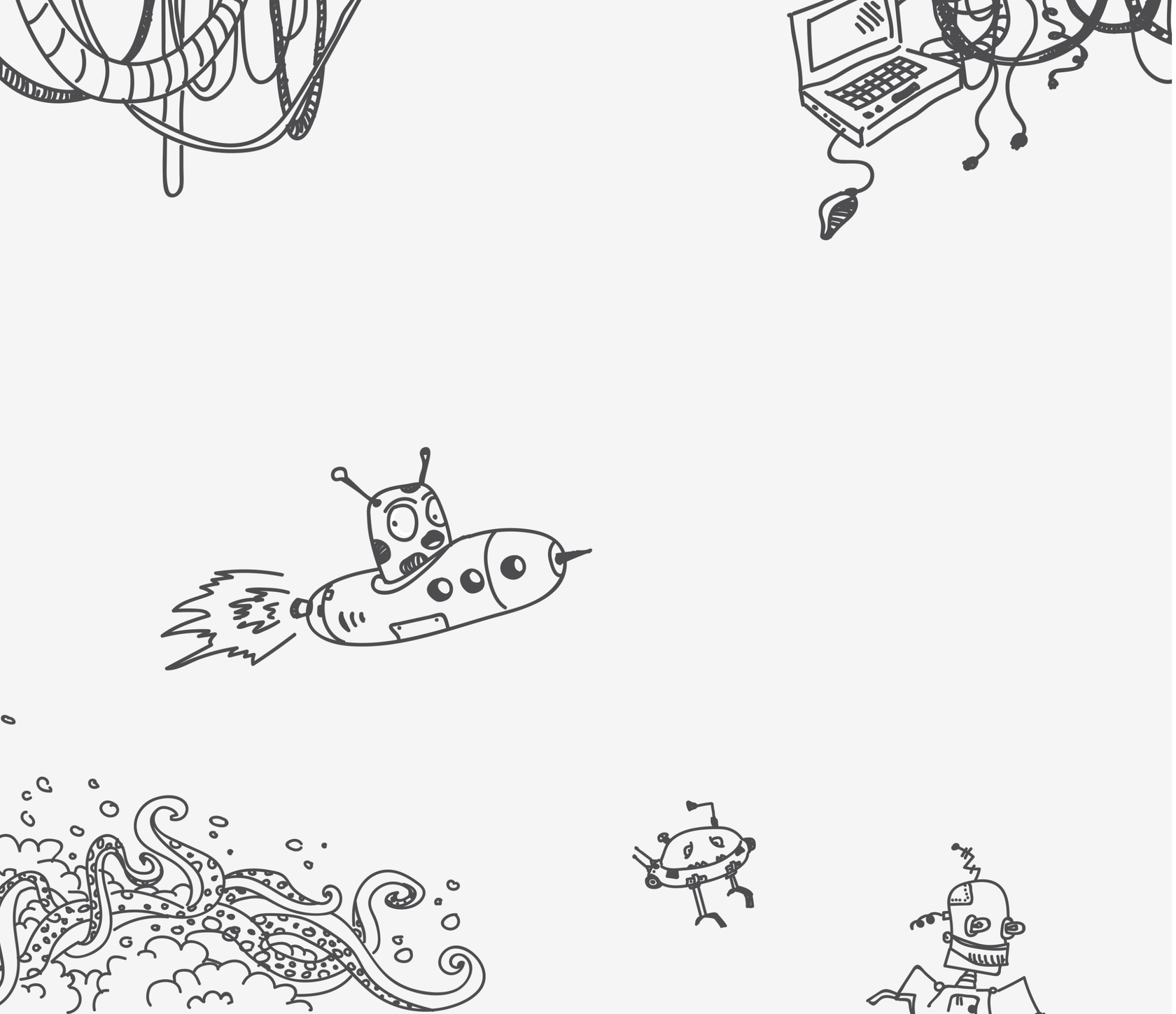
Aim of the analysis

Understanding the data at our disposal

- **acousticness** - a metric indicating how acoustic the track is. Predicts whether a song contains no vocal content. Sounds like "ooh" and "aah" are considered instrumental sounds in this context. Higher values suggest the track is more likely to be instrumental.
- **liveness** - indicates whether the song was recorded with a live audience. Higher values indicate a greater likelihood of a live recording.
- **valence** - describes the positive/negative character of the song. High values of this parameter indicate cheerful and euphoric songs, while low values indicate songs with negative tones (sad, depressing, expressing feelings of anger).
- **tempo** - defines the song tempo in *beats per minute*, BPM
- **genre** - type of song

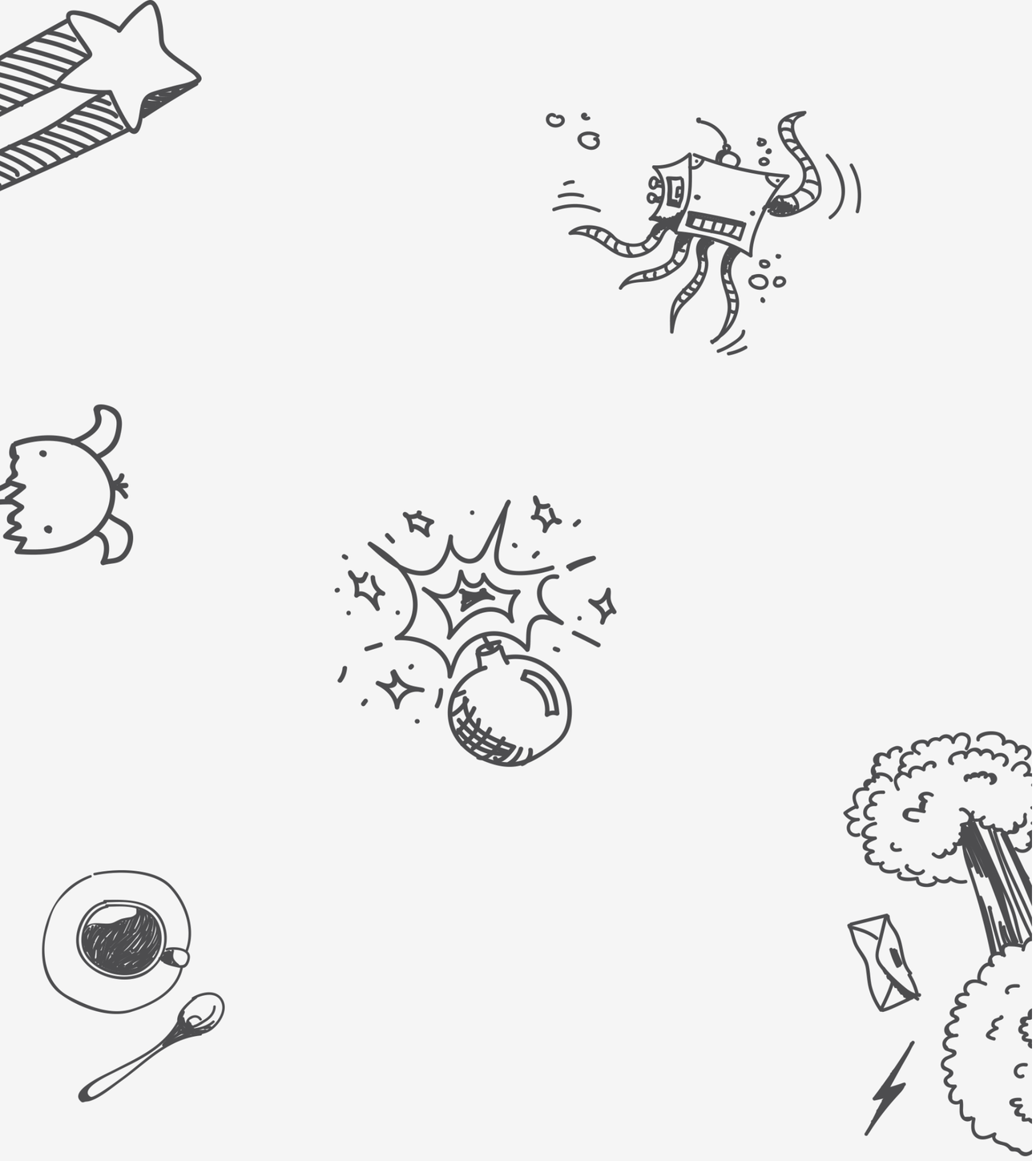


Summary

Summary

Defining and understanding the problem is the most important step in any data analysis. During this step, we determine the scope of our analysis and clarify the objectives based on the perspective of the clients who have requested the analysis.

In this session, we will analyze data on the most popular songs available on Spotify between 2000 and 2019. We've defined the problem and, as diligent analysts, formulated questions to enhance our understanding of the task. It appears there is an additional request regarding the data that will be presented to stakeholders at Spotify's annual corporate meeting. Your task will be to think about what information would be valuable and engaging to the analysis audience.



Practice time

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